

Final Exam 2026: Part 2

GFS189, RTL959

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1 Exercise 1: Volatility Signature Plot and Sampling Choice

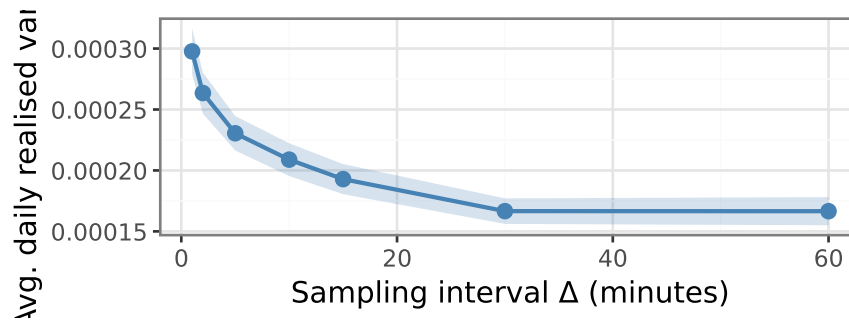


Figure 1: Volatility signature plot for AAPL. Average daily RV ± 2 SE. Bias at $\Delta = 1$ min reflects microstructure noise; stabilises at $\Delta = 5$ min.

The signature plot (Figure 1) shows the characteristic upward bias of the **realized variance (RV)** estimator at $\Delta = 1$ minute, where bid-ask bounce inflates squared log-returns above the true integrated variance (Andersen and Bollerslev, 1998). The curve stabilises by $\Delta = 5$ minutes, motivating our sampling choice throughout; missing minutes are filled by **previous-tick interpolation**. We adopt 5-minute sparse sampling over noise-robust alternatives (TSRV, realized kernels) because the bias is already negligible at this frequency for DJIA stocks and the data lack the tick-level timestamps needed for refresh-time synchronisation.

2 Exercise 2: Daily Realized Covariance

The **realized covariance (RC)** estimator $RC_d = \sum_{k=1}^K r_k r'_k$ sums outer products of synchronised 5-minute log-return vectors (Andersen and Bollerslev, 1998; Barndorff-Nielsen et al., 2011). Intraday gaps are filled by previous-tick interpolation; any remaining missing returns are set to zero, a conservative choice that avoids inflating RC_d .

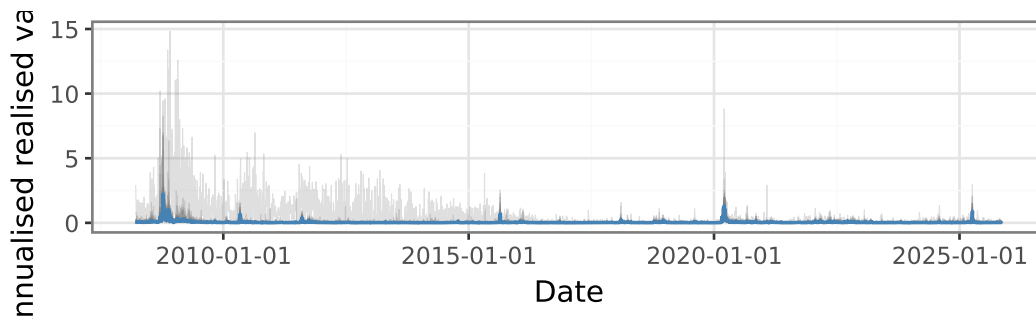


Figure 2: Annualised daily realised variance for all 30 DJIA stocks (grey), cross-sectional median (blue) and IQR (band). Spikes: 2008–09, March 2020, 2022.

The diagonal of RC_d (Figure 2) displays volatility clustering and persistent cross-sectional heterogeneity. The three largest spikes correspond to the 2008–09 crisis, the March 2020 COVID shock, and the 2022 rate-hike episode. At one-minute sampling, off-diagonal elements would be biased toward zero by the **Epps effect** (Barndorff-Nielsen et al., 2011); 5-minute sampling largely eliminates this.

3 Exercise 3: Covariance Estimators

The evaluation window runs from January 17, 2017 to November 14, 2025. (a) **HF random walk:** $\hat{\Sigma}_t^{HF} = RC_{t-1}$ is a martingale baseline — natural when intraday data are available. (b) **LW on daily returns:** $W = 252$ days gives $T/N \approx 8.4$, ensuring a full-rank sample covariance; Ledoit and Wolf (2003) shrinkage toward a scaled identity stabilises the precision matrix for portfolio inversion.

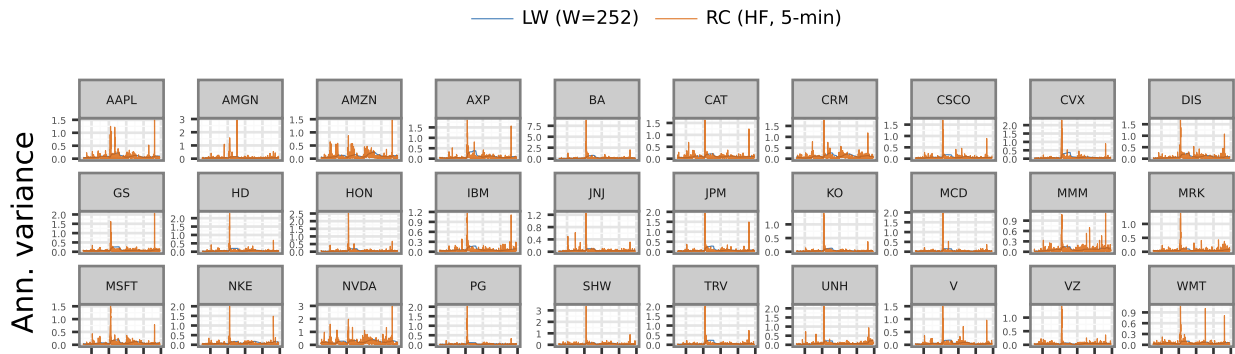


Figure 3: Per-stock annualised variance for all 30 DJIA stocks. LW (orange, rolling 252-day shrinkage) is consistently smoother; RC (blue, 5-min) reacts immediately to volatility bursts.

Figure 3 confirms the smoothing property of LW: the orange line is consistently flatter than the blue (RC) line, which spikes sharply during stress events. The free y-axis reveals cross-sectional heterogeneity — high-volatility stocks (e.g., NVDA, BA) spike far more than defensive names (e.g., KO, WMT).

4 Exercise 4: Minimum-Variance Portfolio Weights

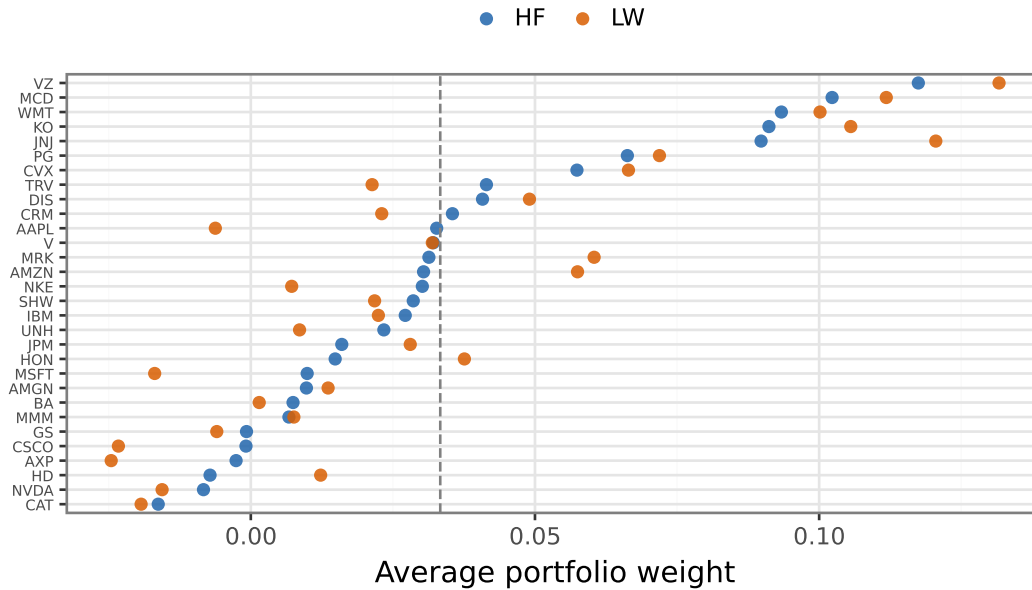


Figure 4: Time-averaged MV portfolio weights for 30 DJIA stocks, ordered by HF weight. Dashed line: equal weight ($1/30$).

The HF strategy assigns its largest average weight to **VZ** (0.117) and its most negative to **CAT** (-0.016). The LW strategy is most positive on **VZ** (0.132) and most negative on **AXP** (-0.025). Average daily turnover is **3.476** (HF) and **0.071** (LW), with the HF strategy showing higher turnover.

Both strategies hold extreme positions — a known consequence of unconstrained MV optimisation ([Ledoit and Wolf, 2003](#)), where small estimation errors cause large weight oscillations.

5 Exercise 5: Out-of-Sample Portfolio Evaluation

Table 1: Out-of-sample performance (second half of sample). Ann. volatility = s.d. $\times \sqrt{252}$; Sharpe ratio = ann. mean / ann. s.d. (risk-free rate = 0, no transaction costs).

Out-of-sample portfolio performance		
Strategy	Ann. Volatility	Ann. Sharpe
HF (RC, t-1)	0.167	0.466
LW (W = 252)	0.141	0.674
Equal-weighted (1/N)	0.179	0.763

Equal-weighted achieves the highest annualised Sharpe ratio (0.763); **HF (RC, t-1)** produces the lowest (0.466). The HF estimator does not outperform LW on Sharpe ratio (0.466 vs. 0.674).

The random-walk estimator uses only a single day per forecast; in the post-2016 window, LW's smoothing advantage dominates, producing more stable weights and a higher Sharpe ratio. HF covariance information adds most value when $N \gg 30$ and the true covariance shifts rapidly ([Ledoit and Wolf, 2004](#)); with only 30 DJIA stocks, LW already captures the dominant structure.

6 References

- Andersen, T. G. and Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4):885–905.
- Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., and Shephard, N. (2011). Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2):149–169.
- Ledoit, O. and Wolf, M. (2003). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance*, 10(5):603–621.
- Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.